

RAD-SMART

The right patient, on the right machine, at a predictable time.

Radiotherapy AI-assisted Dynamic Scheduling, Machine Allocation and Resource Tracking



One planned treatment day: each block is a patient's session (routine, new start, complex, blood irradiation)

Dr Akshay Dinesan, Manipal: team leader and doctor partner · **Abhinand T M**: technical lead

Team: **P S Raghotham Rao · Dr Shirley Lewis Salins · Dr Umesh Velu**

Assistive, not diagnostic

THE PROBLEM

Every weekday, 70–90 patients wait for one machine

Appointment times are given by hand, without knowing how many machine-minutes each patient needs. Too many people are told to come at once, and they wait, every day of a 5–7 week course, usually with a caregiver.

OUR DEPARTMENT'S RECORDS · 55 DAYS, OCT–DEC 2024 · 3,137 PATIENT-DAYS

109 min

one patient in ten waits longer than this, from arrival to treatment

22%

treated more than an hour after their appointment; only 29% within 15 minutes

19 → 66 min

median wait, arriving before 10:00 against after 19:00: delays build all day

The load is rising, from 54 patients a weekday in October to 67 in December, and busy days are worse. 69% of India's 554 radiotherapy facilities have only one machine.

Allocation under hard rules and soft priorities

Hard rules: never broken

- New starts finish by 17:00
- SBRT, SRT, TBI, CSI in 10:00–17:00
- Blood irradiation 13:30–14:00, daily
- 1 breast board, 1 ABC, shared with CT
- MHRC: 16:00–17:00, then the 17:00 bus
- Public transport home by 21:00
- Urgent starts treated by 18:00

Soft priorities: traded off

- Paying patients' preferred times
- Older patients earlier
- Ward, dormitory, nearby: late evening
- Complex cases in 12:00–13:30
- A day of 08:30 to 01:00, little overtime

And every day brings events: machine faults, overruns, no-shows, urgent add-ons and imaging holds, each of which should trigger a re-plan.

FIVE APPROACHES

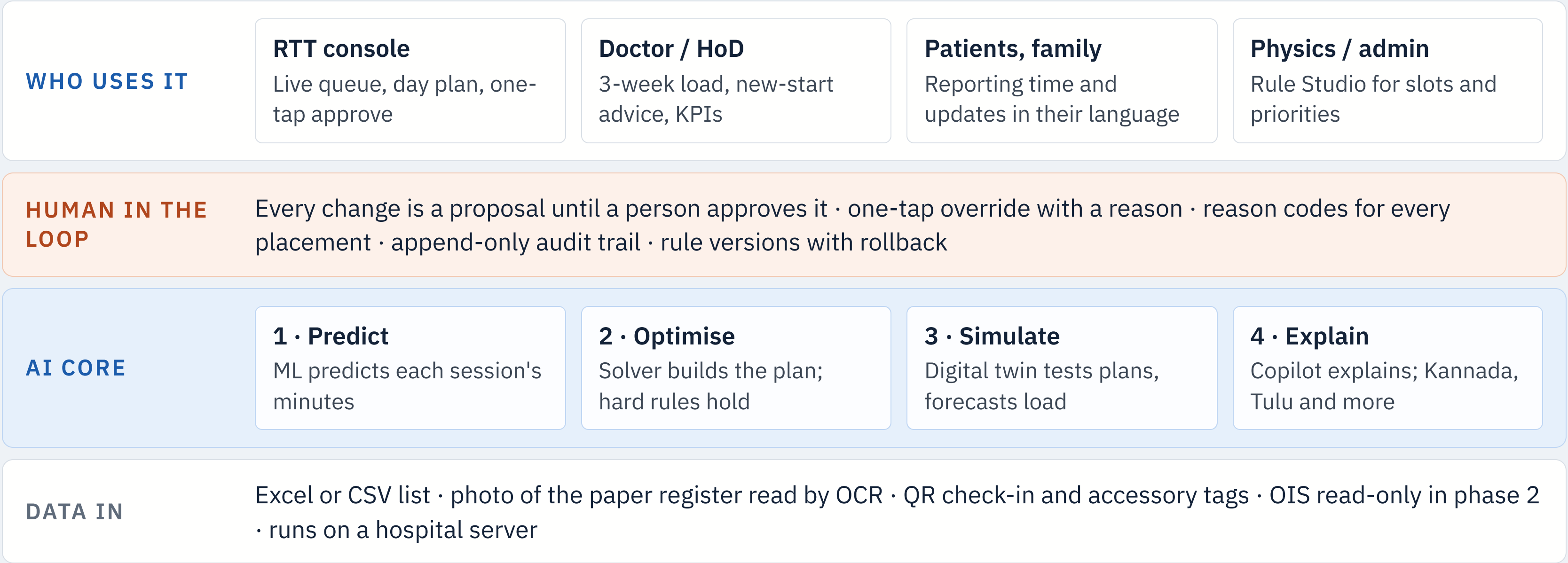
We compared five ways to build it

Approach	Strength	Weakness	Score
Rule-based slot templates	Simple, transparent	Can't absorb variation or faults	3.10
LLM-only agent	Conversational	No guarantee on hard rules	2.60
Static optimisation	Guarantees hard rules	No learning or live recovery	3.60
Deep reinforcement learning	Adaptive in principle	Opaque; needs huge data	2.45
Hybrid (selected): predict, optimise, simulate, explain	Best of each technique	More parts, built in layers	4.85

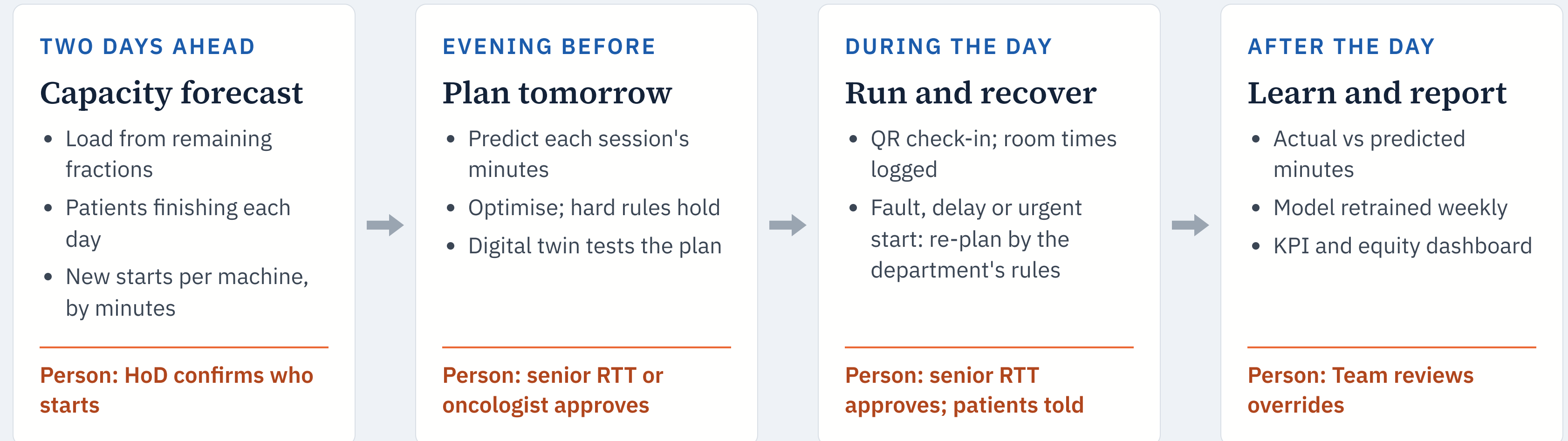
Weighted score out of 5: safety and scope 20%, impact 20%, real-time 15%, feasibility 15%, explainability 10%, ease and languages 10%, configurability 10%.

THE SOLUTION

How RAD-SMART works



Automation proposes; people decide



Every override and actual timing becomes training data, so plans improve each week.

AI where each technique is strongest

PREDICT · MACHINE LEARNING

Minutes, not head-counts

Gradient boosting gives expected and cautious (P80) minutes per session, after starting from the department's estimate table.

PoC: 2.1 min error vs 2.6 for averages

OPTIMISE · OPERATIONS RESEARCH

Hard rules guaranteed

OR-Tools CP-SAT (HiGHS in the PoC) builds and repairs the plan; hard rules cannot break.

PoC: 87 patients, all 7 rules re-checked

SIMULATE · DIGITAL TWIN

Tested before it is used

A Monte Carlo day model, calibrated on the department's records, stress-tests plans and forecasts three weeks of load.

PoC: 300 simulated days per policy

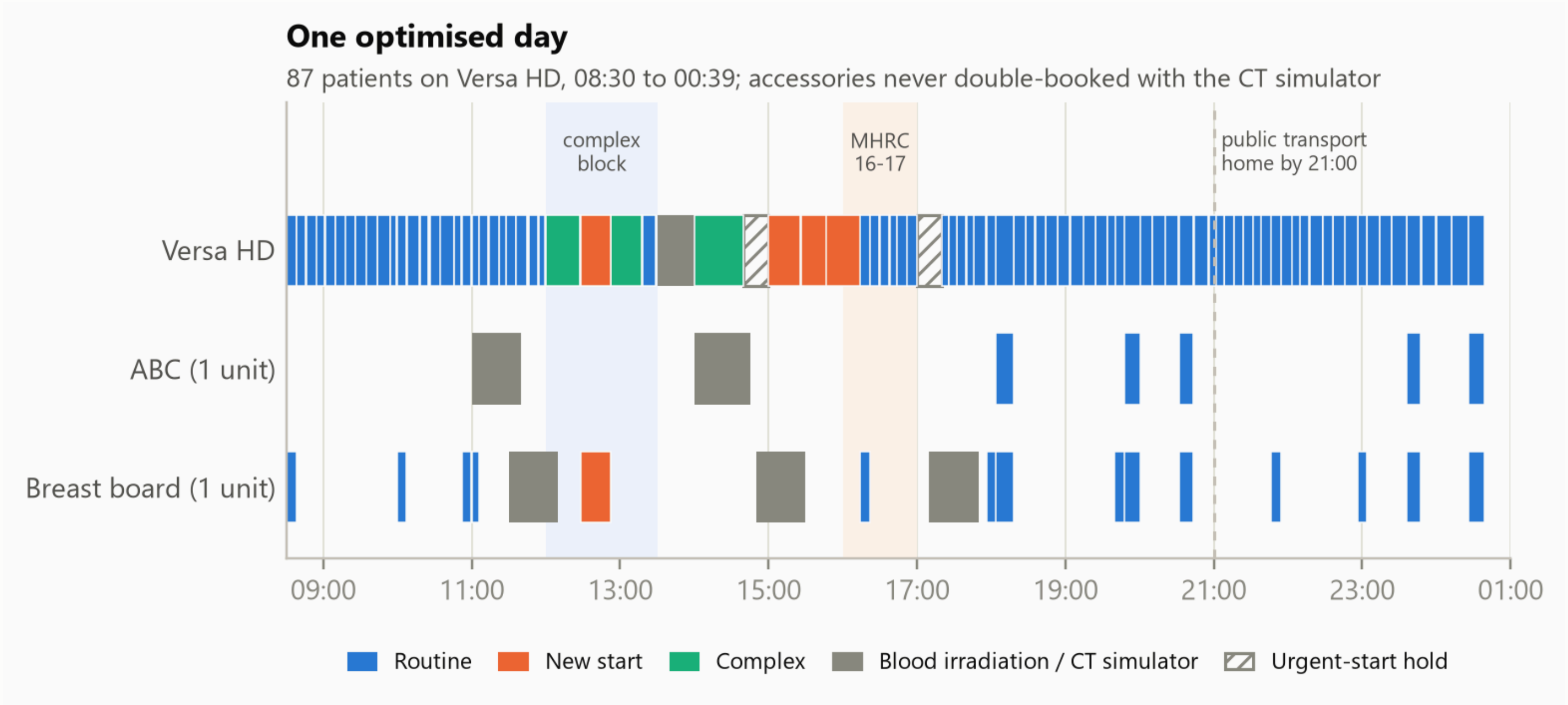
EXPLAIN · SARVAM AI

Kannada, Tulu and more

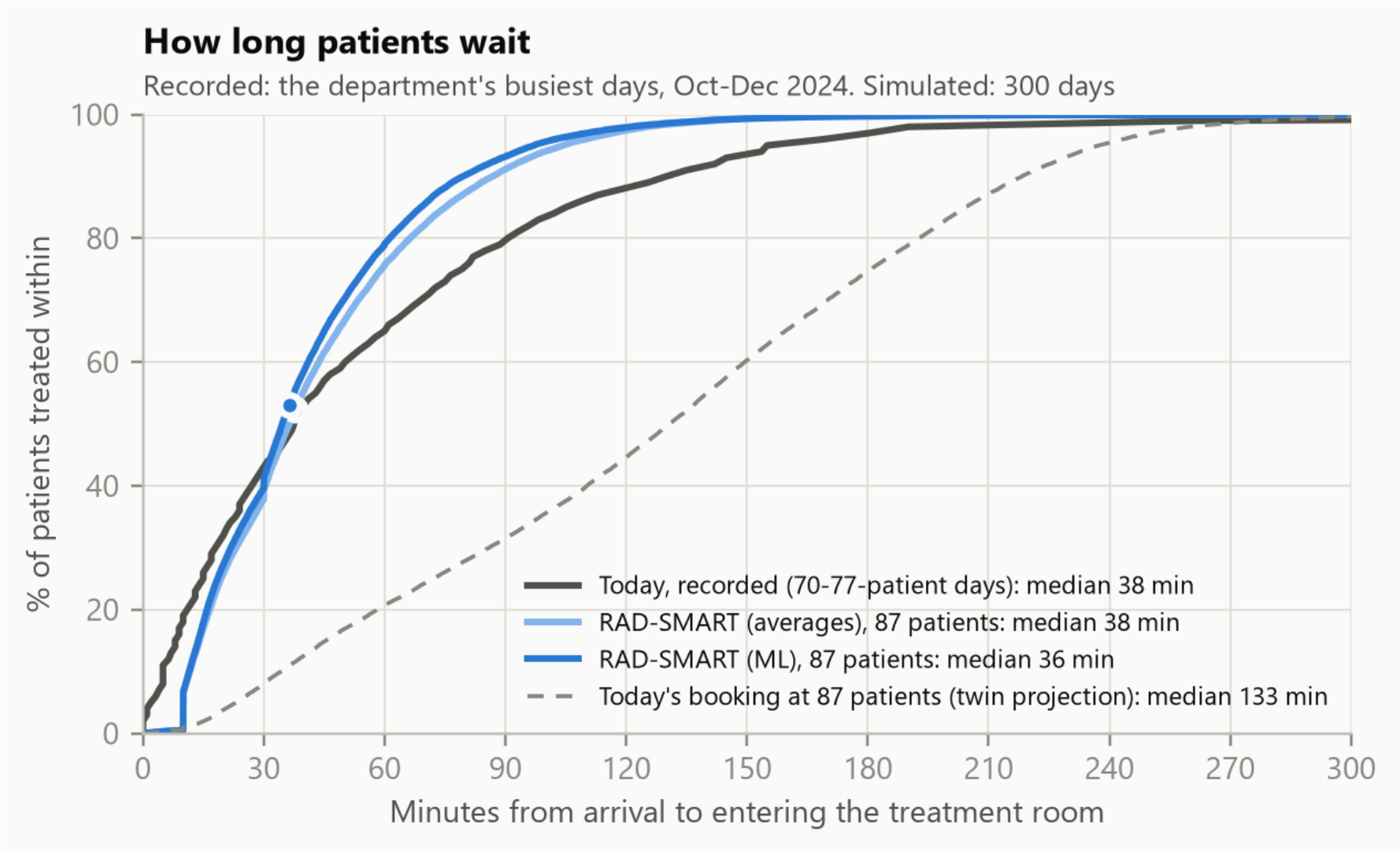
Sarvam-105B explains and edits rules; Bulbul and Saaras reach patients. Tulu: recorded voice notes.

The LLM never schedules

One optimised day, with every rule held



Fewer long waits, and times that hold



130 → 76 min

90th-percentile wait, against the busiest recorded days

27% → 6%

treated more than an hour after their time

30% → 53%

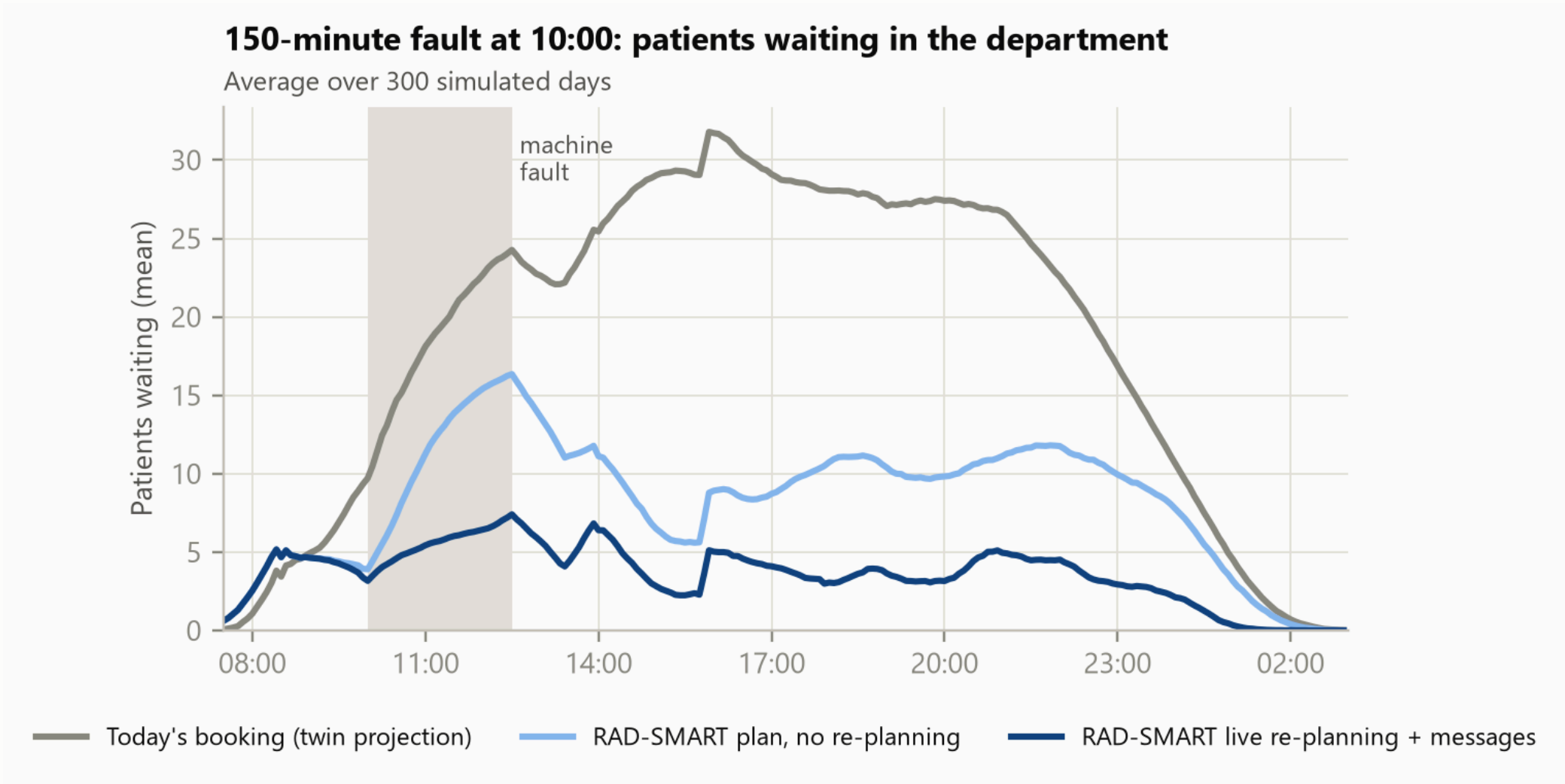
treated within ± 15 min of their time

66 → 36 min

median wait for evening arrivals: no build-up

The typical wait stays about the same (38 → 36 min), because every patient is asked to arrive 20 minutes early to prepare, 45 for pelvic patients. What goes is the long, unpredictable wait.

It recovers by the department's own downtime rules



104 → 36 min

150-min fault: median wait without → with re-planning; 5 new starts to the next day

53 → 39 min

45-min fault: every ongoing patient treated today; 1 new start proposed for the next day

2–11 s

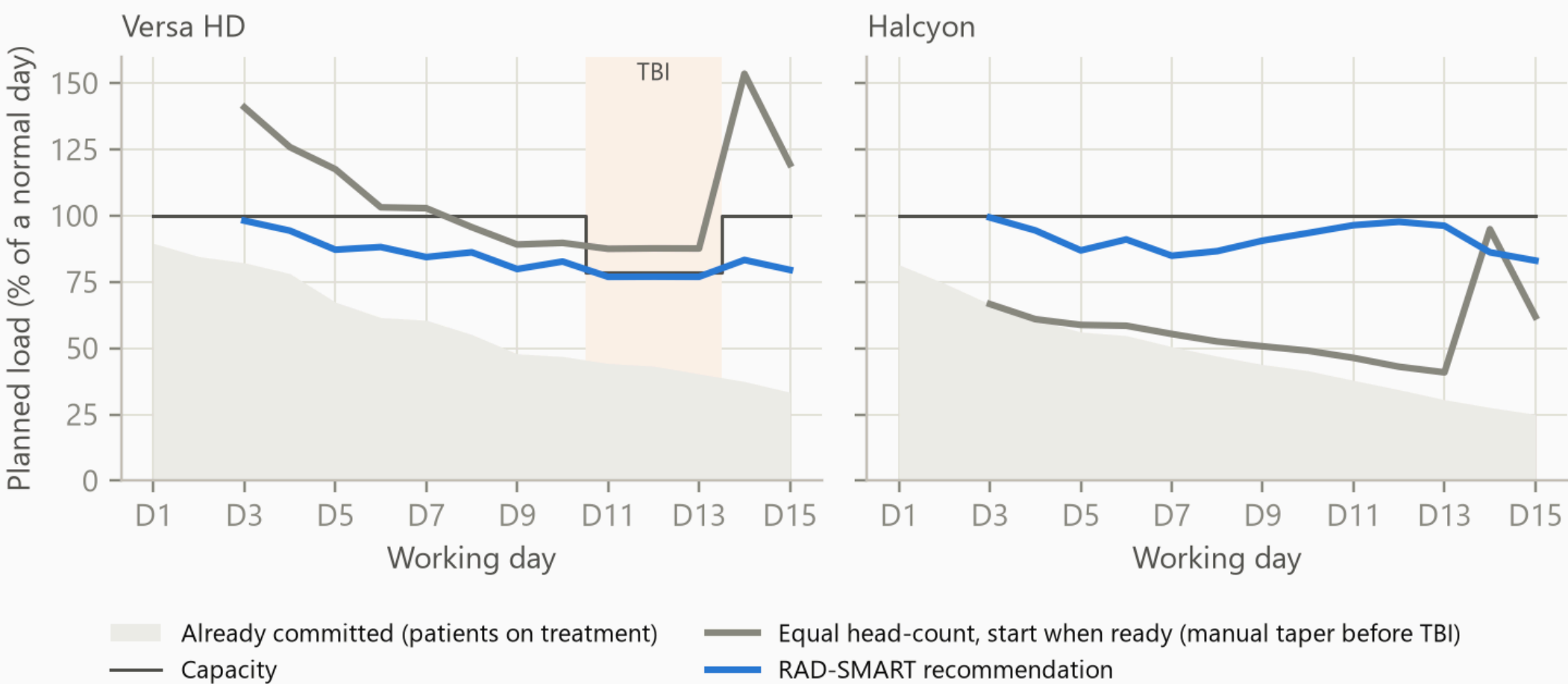
to re-plan the rest of the day

Waiting first

patients already here go first; others are messaged

New starts balanced by minutes, not head-count

Three-week machine-load forecast with new-start recommendations



154% → 98%

peak planned load on the Versa HD

111% → 98%

Versa HD on the TBI days, of the reduced capacity

10 → 0

decision days over capacity on the Versa HD

1.7 → 0 days

from plan ready to first fraction; advice two days ahead

GUARDRAILS

Safe by design

Proposal until approved

A senior RTT or oncologist approves every plan, re-plan and deferral; one-tap override.

Rules re-checked

An independent checker verifies every hard rule before a plan is published.

Clinicians decide

No diagnoses, doses, images or notes are read. Urgency and eligibility come from clinicians.

No identifiers to AI

Models see pseudonymous IDs; names and numbers are added on the hospital server.

Full audit trail

Every plan, override, rule version, model version and message is logged.

Works where the clinic is

Excel, paper via OCR, QR and WhatsApp; runs on the LAN with a printed fallback.

PILOT

90 days on the Versa HD, one primary KPI

Days 1–14

Baseline

Room entry and exit taps added; scheduling unchanged

Days 15–35

Shadow mode

RAD-SMART plans nightly; RTTs compare

Days 36–90

Live

Senior RTT or oncologist approves each plan; patients get times and live updates; model retrained weekly

PRIMARY KPI

$\geq 25\%$ lower

90th-percentile wait from arrival to linac entry (records: 109 min)

ALSO TRACKED

- Over an hour late: 22% → 10% or fewer
- At least 45% within ± 15 min (today 29%)
- Median wait not above baseline; hard rules kept
- Overtime and throughput no worse; no patient group worse off

From prototype to pilot

Team

- Dr Akshay Dinesan (Manipal): team leader, doctor partner and problem owner
- Abhinand T M: technical lead
- P S Raghotham Rao
- Dr Shirley Lewis Salins
- Dr Umesh Velu

Roadmap

- 5 Oct – 8 Nov: build sprint
- 28 Nov: finale, IIT Bombay
- Nov – Jan: approvals
- Q1 2027: 90-day pilot
- 2027: second machine, OIS link, NCG centres

Our ask

- A pilot slot with mentor support
- Sarvam API credits for voice and translation
- Introductions to NCG centres for scale-up

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